



STI CAPSTONE PROJECT MANUAL

General Guidelines

The STI Capstone Project is an undergraduate requirement that allows students to practice various theoretical and practical aspects of their program. It is also a terminal requirement for graduation. Therefore, guidelines and procedures are provided to ensure the Capstone Project 1 and 2 courses are properly implemented.

Formation of Capstone Project Groups

A Capstone Project Group shall comprise a minimum of two (2) and a maximum of five (5) members depending on the complexity of the research/project. A group with more than five (5) members will not be allowed.

Types of Capstone Projects

1. Research Project

Research is a form of systematic inquiry that contributes to knowledge or reaches a new understanding of various phenomena. Through systematic inquiry, the groups will research, collect, organize, and analyze information from primary and secondary sources to address a gap or problem. *Refer to the STI Research Writing Manual.*

2. Externally-Oriented Projects

The Capstone Project Group develops a solution for a client. The client must be an actual organization with existing issues to be resolved. It focuses on the infrastructure, application, or business processes in implementing a computing solution to a retail problem. The groups are provided context, scenario, or problem and work toward a solution, which can be a product, software, system, and technology.

Securing a Client

The most straightforward step is to start with who your group already knows. Ask the organization where you had your on-the-job training, connect with friends, family, and your school's alumni, and ask for referrals. Most importantly, coordinate with your Capstone Project Coordinator to assist you in securing a client. A client can be in any of the following groups:

- Existing or start-up businesses
- Private/non-government organizations
- Local government units (LGU)
- Government departments/agencies
- Schools
- Specific groups of people

Although you will receive many referrals, it is helpful to be guided by the summary checklist below (Atkinson & Dong, n.d).

- Does the potential client have an active interest in the project?
- Are they willing to work with your group for two (2) semesters?
- Is the company willing to share confidential information with the group?

- Will the company give you access to the right people?
- Is the company open to using information technology to improve its operations?

During initial interactions with the potential client, the group must be wary of these red flag signs before continuing with the project:

- The client has unrealistic expectations from the group; does not consider time, skills, and competency constraints.
- The client is not flexible and insist on a specific solution/system/approach.
- The client who requires a group to develop an overly comprehensive plan without providing any guidance.

Research/Capstone Project Topic

The Capstone Project Group must choose their topic carefully. The topic must provide an opportunity for the Capstone Project Group to apply knowledge and skills learned throughout their studies to solve a real-world problem in their field and demonstrate proficiency in relevant research.

The Capstone Project Group's ability to find financial and time resources to research the topic will also affect their capability. Some topics are unlikely to be completed in the time allowed by the course of study. This may be because some require measuring an intervention's impact over a long period.

Capability also means the Capstone Project Group must be reasonably confident of gaining access to any data they might need to collect. Many people start with ideas where access to data will prove difficult. Specifically, more sensitive topics, such as customer satisfaction or decision-making by senior managers, are potentially fascinating. However, they may present many access restrictions.

If the Capstone Project Group has chosen an idea or topic, they must ensure that the question(s), aims, and objectives relate clearly to the idea.

Although the topic may depend on the needs of the client or organization, it is helpful to check and continue to check any potential topic against the summary checklist below.

Attributes of a Good Capstone Topic
<i>Capability: is it feasible?</i>
Is the topic something with which you are fascinated?
Is the topic achievable within the available time?
Will the topic still be current when you finish your project?
Is the topic achievable within the financial resources that are likely to be available?
Are you reasonably confident of being able to gain access to data you are likely to require for this topic?
<i>Benefits: is it useful?</i>
Will the proposed topic improve the client's operations and current processes?
Will your proposed project be able to provide fresh insights into the industry as a whole?
<i>Social responsibility: is it relevant?</i>
Is your topic socially relevant?
Does your topic address a specific social or organizational problem?
Will your proposed project provide practical applications to society not just to the organization?

Capstone Project Writing

A template for the different parts of the document will be available in the STI eLMS to ensure uniformity of appearance and required contents.

The following discusses the general mechanics of writing the capstone documentation:

1. Appearance
 - a. The manuscript must be prepared using a computer.
 - b. Unnecessary markings and corrections are not accepted.
 - c. Printing the original copy should be letter quality with dark black characters consistently clear and dense. Dot-matrix is unacceptable.
 - d. The manuscript shall be printed on only one (1) side of each page.
2. Paper
 - a. White bond paper
 - b. Letter (8.5" × 11") paper size
 - c. Substance 20 or higher
3. Margins and Footer
 - a. Margins must be uniform on all pages. Margin requirements apply to all materials included in the research, such as figures, tables, photographs, etc., and all materials in appendices.
 - b. Page numbers must be at the bottom of the page, right side. The school's name must be at the bottom of the page, left side.
 - c. The margins shall be 1.5 inches on the left and one (1) inch at the top, bottom, and right sides.
4. Font
 - a. The official font typeface is Times New Roman. The official font size is 12.
 - b. The official font typeface and size must be used for all materials in the text.
 - c. Italics should be used only to highlight specific words or phrases in the text.
5. Spacing
 - a. 1.5 line spacing is required throughout the manuscript except for bibliographic entries and references, long-quoted passages, figure and table captions, and items in the lists and tables, which should be single-spaced.
 - b. Preliminary pages such as the Title Page, Abstract, Approval Sheet, and Table of Contents have special spacing requirements. Refer to the *Capstone Manuscript Template*.
6. Text Alignment
 - a. The body of the text must be (fully) justified.
 - b. The first line of each paragraph shall be indicated by a tab indentation (0.5 inch) throughout the document.
7. Manual Style
 - a. It is recommended that the Publication Manual of the American Psychological Association, 7th edition (2020), be used in relation to formatting in-text citations, direct quotations, and references.
8. Content and Sequence of Sections

The capstone manuscript must have the following content and sequence of sections:

Title Page	
Endorsement Form for Final Defense	
Approval Sheet	
Acknowledgments	
Abstract	
Table of Contents	
List of Tables	
List of Figures	
List of Abbreviations	
Chapter I: Introduction	
A. Project Context	
B. Capstone Project Questions	
C. Significance of the Project	
D. Scope, Limitations, and Delimitations of the Project	
E. Review of Related Literature	
F. Theoretical/Conceptual Framework	
Chapter II: Methodology	
Chapter III: Results	
Chapter IV: Discussion	
References	
Appendices	

Writing the Title

Your title summarizes the main idea of your capstone project. As such, it should not be too long with unnecessary words or too short, making it too general. You may first provide a draft title at the start of your project and then write the final title after writing your paper.

Below are some characteristics of effective capstone titles (USC Libraries, 2022):

- The subject and scope of the project are indicated in the title.
- Capitalize all the first words, including the first word of the subtitle. All nouns, pronouns, verbs, adjectives, and adverbs that appear between the first and last words of the title are capitalized.
- Lowercase only minor words that are three (3) letters or fewer in a title or heading (except the first word in a title or subtitle or the first word after a colon, em dash, or end punctuation in a heading):
 - short conjunctions (e.g., "and," "as," "but," "for," "if," "nor," "or," "so," "yet")
 - articles ("a," "an," "the")
 - short prepositions (e.g., "as," "at," "by," "for," "in," "of," "off," "on," "per," "to," "up," "via")
- Use words that create a positive impression and stimulate reader interest.

The following criteria should be satisfied in evaluating a capstone project title:

- a. *Conciseness*: The title should reflect the key ideas expected in the outcome so that readers can easily understand the nature of the project.
- b. *Reflective of the content, problem, and main variables of the study*: The title should be narrowly focused and carefully defined, covering a broad-ranging, complex research problem.
- c. *Free of value judgment*: The project is not a fact nor a question but what the Capstone Project Group aims to develop or prove.

Examples of Capstone Project Titles:

- Designing Mobile Technologies for Environmental Sustainability of ABS Consulting Group
- Evaluate and Redesign Asian Airlines Website
- Ensuring Food Safety and Improving Service Delivery of Country Style Food Corporation Through the Use of an Online-Based Food Evaluation Platform
- Increase Customer Engagement in Buzz Store through the Touchscreen Based Point-Of-Sale System (POS)
- Package Design for Art Box's Graffiti Street-Ready Spray Paints

Writing the Introduction

The introduction includes what your project intends to address and provides an overview of your work. It usually consists of the following:

A. Project Context

It presents the 'big picture,' identifying the context of the problem to be investigated. It sets the stage for the relevance and purpose of the study. A description of the organization's existing conditions and prevailing situations can be included here. It should include sufficient background information about the project.

B. Capstone Project Questions

The capstone questions provide direction to the project and writing process. They must be relevant to the studied discipline and actual client. They must explore any uncertainty in an area of concern that needs investigation. The questions should be specific and clearly indicate how they might be answered or accomplished.

Example of Capstone Project Questions:

1. What are the perceptions of Asian Airlines' customers about the color and design of their website?
2. What is the actual number of Buzz Store's customers, per demographic, who answered the feedback form through the store's tablet?

C. Objectives of the Project

It describes the purpose of the capstone project. It aims to answer these questions:

- How will the project be important, and how will it help improve practice? Who will it help, and how? Will it provide a stepping-stone for others to go further?
- What benefits would occur if definitive answers were found to the capstone project questions?
- How will the client benefit from the capstone project?

D. Scope, Limitations, and Delimitations of the Project

The *scope* refers to the extent to which the project will be completed. It must include what the project will cover and the factors within its acceptable range. For example, if you evaluate the perceptions of Asian Airline's customers on its marketing materials, the scope could include the company's official website and social media accounts.

Limitations are restrictions on the extent or type of available data. Limitations are occurrences that are out of the Capstone Project Group's control and may restrict the generalizability of findings. Limitations may include that the Capstone Project Group was given access only to certain information or people in the organization.

Delimitations are the boundaries set by the Capstone Project Group. They include factors or variables not included in the project's conduct. Delimitations may include the period covered by the project, the location where the project took place, and the population included in the project, among other things.

E. **Review of Related Literature**

It conveys to the reader the knowledge and ideas established about the topic from published and unpublished materials. It collects critical sources about your chosen topic. For example, suppose your project is on how store design affects customer satisfaction. In that case, you will need to provide an overview of the principles of store design layout, customer satisfaction, and prior research on said topics.

When choosing references, it is best to use those published within the last five (5) years from the date of writing the paper. However, the relevance of sources varies depending on the project topic. Foundational works are worth citing to establish the project context. It is recommended that undergraduates have at least ten (10) references for their papers.

When writing a review of literature, keep these issues in mind:

1. *Use evidence:* Your interpretation of the available sources must be backed up with evidence (citations) that demonstrates that what you say is valid.
2. *Be selective:* Select only the most important points in each source to highlight in the review. The type of information you choose to mention should relate directly to the research problem.
3. *Use quotes sparingly:* Some short quotes are okay if you want to emphasize a point or an author's statement cannot be easily paraphrased. Sometimes you may need to quote certain terminologies coined by the author, uncommon knowledge, or words taken directly from the study. Do not use extensive quotes as a substitute for your summary and interpretation of the literature.
4. *Summarize and synthesize:* Remember to summarize and synthesize your sources within each thematic paragraph and throughout the review. Synthesize the important features of the study by rephrasing the study's significance and relating it to your work.
5. *Keep your voice:* While the literature review presents others' ideas, your voice (the researcher's) should remain front and center. For example, weave references to other sources into your writing but maintain your voice by starting and ending the paragraph with your ideas and wording.
6. *Use caution when paraphrasing:* When paraphrasing a source that is not your own, be sure to accurately represent the author's information or opinions in your own words. Even when paraphrasing an author's work, you still must provide a citation to that work.

Below are the main types of literature that will most likely be used in research (Saunders & Philips, 2018):

- *Textbooks* – These are written specifically for students or professionals. It is useful as an introductory source for an overview of the research topic and to find recognized experts.
- *Peer-reviewed academic journal articles* – These provide detailed reports of research. Articles are written by experts and evaluated by other academics (peer reviewers) to assess their quality and suitability.
- *Non-refereed academic journal articles* – These are articles selected by an editor or editorial board with subject knowledge.

- *Professional and trade journal articles* – These are written for professional or business organization members. These can provide valuable insights that can be put into practice, although they may be biased.
- *Newspaper articles*—These are articles written for members of the public. The news presented is filtered dependent on events, with priority given to headline-grabbing stories that are likely to appeal to that newspaper’s readers. Newspapers are a good source of topical events and developments.
- *Conference proceedings* – These are selected papers presented at a conference, often published online, as a book, or special edition of a journal. These are usually peer-reviewed.
- *Reports* – These are reports on specific topics written by academics and various organizations, including market research organizations and government departments.
- *Theses and dissertations* – These are research papers written by students as a requirement to complete undergraduate, master’s, or doctoral programs. They demonstrate a student’s learned knowledge and enhance society by advancing knowledge.
- *Existing systems* – These may be included if they relate to the chosen topic and are relevant to the project.

Plagiarism is when someone presents the work and ideas of other people as if they were their own without acknowledging and referencing the source (Saunders & Philips, 2018).

To prevent this, groups must record the sources of all the information, research findings, theories, and other ideas. This process is called **referencing**, which allows the researcher to acknowledge and give credit to the work of others. Referencing is essential for the following reasons (Altinay & Paraskevas, 2008):

- It shows that the group has completed the recommended reading independently. The group’s breadth of reading enhances the credibility of their work.
- It shows that the group actively engages in academic debate and positions their ideas with others.
- It helps the group establish a basis for their claims and arguments, thus contributing to their ideas’ credibility.
- It will prevent accusations of plagiarism.

F. **Theoretical/Conceptual Framework**

A theoretical framework limits the scope of the relevant data by focusing on specific variables and defining the viewpoint (framework) the researcher will take in analyzing and interpreting the data to be gathered. The theoretical framework is the theory on which the study is based.

A conceptual framework is a *structure* that the researcher believes can best explain the natural progression of the phenomenon to be studied. It is the researcher’s explanation of how the research problem would be explored. The conceptual framework describes the relationship between the main concepts of a study. A conceptual or schematic diagram is required to visually explain the variables and relationships between them that need to be studied.

Writing the Methodology

This section should describe the process and techniques used to gather, analyze, and process data to investigate and address the problem. It must include how data was obtained as it affects findings and interpretation of results. The following points must be made clear in this section:

1. Sample/population of the study
 - What are the characteristics of the sample/population?

- How did you arrive at the sample size?
 - How did you select the participants?
 - Why were they chosen?
 - How did you collect the samples?
2. Data gathering tools
 - What equipment/tools did you use?
 - What considerations were undertaken to ensure validity and reliability?
 - Are the tools used adapted from another source?
 3. Data gathering procedures
 - What methods (interviews, surveys, focus groups, experiments, program evaluation, etc.) did you use to gather data?
 - How were the methods conducted?
 - How were the data retrieved?
 - How long did the gathering last?
 - How were the data analyzed and summarized?
 - What were the statistical methods or tests used to analyze the results?
 4. Ethical Considerations
 - How did you ensure the confidentiality of information and data privacy of participants?
 - How were the data secured?

The following must be included if the project consists of product, software, or system development:

1. Requirement analysis
 - What are the computing solutions of the proposed software or system that address the needs of the organization?
 - What product features/requirements are included in the proposed output that solves the client's problem?
2. Requirements Documentation
 - What are the product/software/system features agreed upon by the client and the Capstone Project Group?
3. Design of Product, Software, or System
 - How was the proposed product, software, or system designed following the standards of the client?
4. Development and Testing (if applicable)
 - How was the proposed product, software, or system tested or evaluated?

Writing the Results

In this section, the researcher must present the data and information gathered. Use this section to illustrate your data better by showing figures, tables, charts, graphs, etc.. Highlight relevant values in presenting the findings and answering the research questions.

Below are some points to consider:

- Present a profile of the research subjects, including relevant demographics about them.
- Provide well-thought-out tables and diagrams with descriptive texts.
- Summarize transcripts for analysis by providing pieces of evidence through direct quotations from the interviews
- Show the survey questionnaires provided and tabulation of results

Include the following for product, software, or system development:

- Implementation plan on how the product, software, or system will be deployed and installed
- Site-specific implementation requirements/resources
- Implementation results (if applicable) as a ground for improving the project

Writing the Discussion

This section must answer: *What do the data presented mean or imply?* Clear answers to the capstone project questions and problems must be provided in this section. Insights drawn from the project findings must be included to help the reader fully understand your findings. You may show how your results or interpretations agree with other published works. Discuss any correlations, if any, and unresolved or unexpected outcomes.

End the discussion with a closing summary on the significance of the project output. The recommendations must include suggested improvements in how the project was implemented, limitations encountered during the conduct of the study, including future directions that can still be made about the chosen research area. Recommendations may also include further enhancement that can be done to the developed project, software, or system.

Writing the Abstract

This is the last section to be written. It includes an overview and key information about the research:

- Problems/issues that were addressed
- Significance of the project
- Methodology
- Main findings
- Conclusion

Appendices

It may include relevant source codes, evaluation tools and documents, sample input/output reports, and user guides.

Writing the Acknowledgements

It is where you acknowledge the contributions of those who helped or assisted you with your research. Always maintain a formal and professional tone to convey your sincere gratitude to those who contributed significantly to your research.

Other Matters

Submission of Approved Capstone Project Output

Once the defense panel has approved the capstone project, make one (1) copy of the approved project and have it hardbound. The binding should use either a black or dark blue leatherette cover, with the same text of the title page engraved in gold or silver letters on the front cover. The capstone project title, the researchers' names (first and middle initials and full surname), and the year of approval must be engraved in gold or silver letters on the spine.

The copy must be submitted to the Research Coordinator. The research output shall be stored in the school library for five (5) years.

The research group shall also submit a soft copy of the approved research output to the Research Coordinator and save it in the e-Learning Management System.

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